TO BELIEVE OR NOT TO BELIEVE TO YOUR LLM

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ARTIFICIAL INTELLIGENCE

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OVERVIEW

1 Goal

Defining the research goal

2 Background

- Uncertainty
- Heuristic Approaches
- Contribution of the paper

3 Methodology

- Iterative prompting
- Epistemic uncertainty estimation

4 Conclusions

- Threshold estimation
- Experiments
- Future Works

PROBLEM

LLMs can sometimes produce very confident answers that are actually incorrect. **Hallucinations** make it difficult for users to understand when the model's output can be trusted.

GOAL

Quantify the reliability of an LLM's answers.

METHODOLOGY

Understanding how large is the ***epistemic uncertainty***, which is responsible for hallucinations.

UNCERTAINTY

ALEATORIC

Aleatoric uncertainty reflects the **irreducible randomness** in a prediction problem. It arises from noise or natural variability in the data, such as when multiple valid answers exist for the same query.

EPISTEMIC

Epistemic uncertainty arises from the model's **lack of knowledge** about the true answer. This occurs when there is insufficient training data or the model has limited capacity, leaving it unsure about how to respond reliably.

In particular, by examining *epistemic uncertainty*, we can get insights into how **truthful** a model's predictions are.

When epistemic uncertainty is **low**, the model's outputs tend to be **close to the ground truth**.

However, with deep neural networks it is often difficult to accurately measure uncertainty.

We face challenges such as:

- Weak guarantees about how large the variance of predictions is
- Inconsistency: by training more data, the model does not necessarily converge to the true function.

HEURISTIC APPROACHES

1. looking at the log-likelihood of answers of the model
2. Entropy of the outputs: how sparse the distribution is
3. Ensembling: build more models and look at the probability between their predictions
4. Conformal predictions: provides guaranteed predictive ranges

⚠ They are meaningful only where there exists a single correct response

When we are dealing with multiple correct answer, we have aleatoric uncertainty in the ground truth. Therefore, quantifying how much is the total uncertainty is not enough.

⇨ We need a metric that detect only epistemic uncertainty, to detect hallucinations

CONTRIBUTION

The paper aims to:

1. **Differentiate** between **aleatoric** and **epistemic** uncertainty in LLMs answers, allowing to:
   - deal with multiple-response queries
   - discover hallucinations

2. Develop an **iterative prompting** method that generates multiple answers to the same query, in order to evaluate if the responses are independent from each other
   → if answer are dependent the model has high epistemic uncertainty

3. Derive an information-theoretic metric for epistemic uncertainty, which is based on the joint distribution of responses (that is also independent to aleatoric uncertainty)
   → **Mutual Information (MI)**

4. Follow a computable **lower bound** on the previous metric and discuss an algorithm for hallucination detection based on thresholding a finite sample of the MI estimator

5. Focus on a simple mechanistic explanation for explaining how the model output is influenced by previous answers during the iterative prompting

BASIC NOTATION

$\mathcal{Z}$: space of possible texts or answers

$\mathcal{M}_1(\mathcal{Z})$: set of all possible probability distributions we could have over the answer

X : set of finite sentences built over a given alphabet

Q : distribution learned by the LLM, which is how the model generates answers

P : ground-truth conditional probability distribution, which is how the real language would distribute the answers

$Y_1 \dots Y_t$: set of possible answers given a fixed query

$\mathcal{F}_t(x, Y_1, \dots, Y_t)$: prompt generator that, given a query and previous responses, produces a new prompt

INFORMATION-THEORETIC NOTIONS

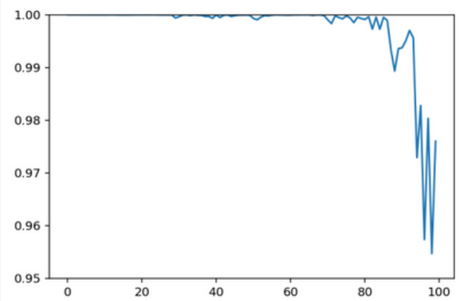
$$D_{KL}(\mu, \mu') = \sum_{z \in Z} \mu(z) \ln\left(\frac{\mu(z)}{\mu'(z)}\right)$$

The **Kullback-Leibler divergence** measures how much two probability distributions differ. When the distributions are similar, the overall divergence becomes very small. If the distributions are different the divergence assumes an high value.

$$I(\mu) = D(\mu, \mu^{\otimes}), \text{ where } \mu^{\otimes}(z) = \prod_{i=1}^{n} \mu_i(z_i)$$

The **Mutual Information** is a special case of the Kullback-Leibler divergence: instead of comparing two arbitrary distributions, it compares a joint distribution with the product distribution of its marginals. It quantifies how much error we make when we treat the variables (strings of text) as independent.

REPETITION EFFECTS



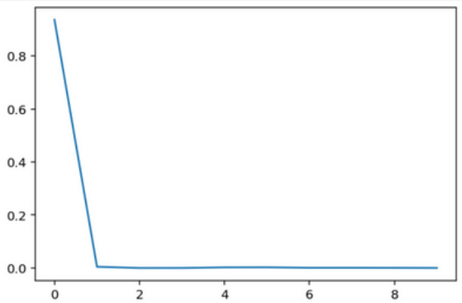
Q: *What is the capital of the UK?* A: *London* ($\approx$ 1.0) and *Paris* (1.29×10^{-10}).

1. Single correct answer

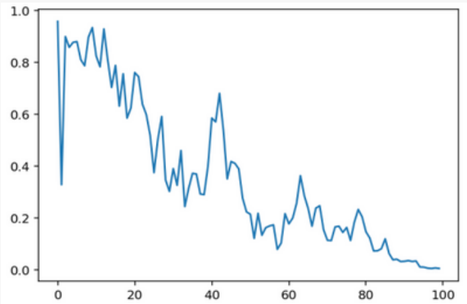
- Low epistemic and aleatoric uncertainty
- Small effect on the conditional normalized probability
- Model appears robust

2. Multiple plausible answers

- High epistemic uncertainty
- Strong effect on the conditional normalized probability
- Model is highly influenceable by repeated in-context information



Q: *What is the national instrument of Ireland?* A: *The harp* (0.936) and *Uilleann pipes* (0.063).



Q: *Name a city in the UK* A: *London* (0.958) and *Manchester* (0.041).

3. Multiple correct answers

- High aleatoric uncertainty
- Moderate effect on the conditional normalized probability, often resulting in oscillations rather than a collapse
- Model recognizes the multiplicity of correct responses

WHY CAN REPEATING AN INCORRECT ANSWER IN THE PROMPT INFLUENCE THE MODEL'S RESPONSE?

→ This phenomenon can already be observed in a single self-attention head.
Its function is:

$$f(\mathbf{Z}; \mathbf{W}^{\mathbf{Q}}, \mathbf{W}^{\mathbf{K}}, \mathbf{W}^{\mathbf{V}}) = \mathrm{Softmax}(\frac{1}{\sqrt{d}} E^{\intercal} \mathbf{W}^{\mathbf{Q}} (\mathbf{Z}\mathbf{W}^{\mathbf{K}})^{\intercal}) \mathbf{Z}\mathbf{W}^{\mathbf{V}}$$

Step 1: Compute similarities between the end-of-input vector $E^{\intercal}\mathbf{W}^{\mathbf{Q}}$ and projections of each row in the input matrix $\mathbf{Z}\mathbf{W}^{\mathbf{K}}$ → produces a vector of scores (one per input row).

Step 2: Apply Softmax to obtain normalized attention weights.

Step 3: Multiply the weights by $\mathbf{Z}\mathbf{W}^{\mathbf{Q}}$ (the values) → weighted sum of input information produces the output.

⇨ The head looks at all sentences in the prompt, assigns a weight to each, and generates the output as a weighted combination of all values.

Repeating a response in the prompt **increases its cumulative weight** in the Softmax.

IN-CONTEXT VS IN-WEIGHT LEARNING

IN-WEIGHT

If the query X appeared many times in training, the model relies on its learned parameters. Output depends mostly on X itself, the true distribution $P(.\,|X)$ is produced.
$\rightarrow$ $E^{\intercal}\mathbf{W^Q}(\mathbf{W^K})^{\intercal}X$ is large, dominating the attention.

IN-CONTEXT

If the query X is rare in training, but a response Y occurs many times in the context Z:
Each occurrence of Y contributes to the attention scores.
The head produces an output copying Y rather than responding based on X.
$\rightarrow$ repeated Y increases its weight in the Softmax $\rightarrow$ output biased toward Y.

IDEAL INDEPENDENCE VS. MODEL DEPENDENCE

GROUND TRUTH ASSUMPTION

The ground-truth satisfies

$$P(Y_t \mid F_{t-1}(x, Y_1, \ldots, Y_{t-1})) = P(Y_t \mid x), \quad \forall\, t \in \mathbb{N},\ x, Y_1, \ldots, Y_t \in X.$$

PSEUDO JOINT DISTRIBUTION

Given a family of prompt functions $\mathcal{F}$, a conditional distribution $\mu \in \mathcal{P}$, and $n \in \mathbb{N}$,

we use notation $\tilde{\cdot}$ to denote a pseudo joint distribution defined as $\tilde{\mu}$ as

$$\tilde{\mu}(Y_1, \ldots, Y_n \mid x) = \mu(Y_1 \mid F_0(x))\, \mu(Y_2 \mid F_1(x, Y_1)) \cdots \mu(Y_n \mid F_{n-1}(x, Y_1, \ldots, Y_{n-1})).$$

THE METRIC

EPISTEMIC UNCERTANTY METRIC

Given an input $x \in X$, we say that the epistemic uncertainty of $\tilde{Q}$ is quantified by $D_{\mathrm{KL}}(\tilde{Q}, \tilde{P})$

THEOREM

For all pseudo joint distributions $\tilde{Q}$ satisfying the ground truth assumption , we have that

$$D_{\mathrm{KL}}(\tilde{Q},\ \tilde{P})\ \geq\ I(\tilde{Q}).$$

THE PROBLEM AND THE SOLUTION

Thanks to the previous theorem I have a metric based only on a **known** distribution?
Actually no: since the support of Q is (potentially) **infinite**, it is impossible to compute the metric over the entire region where it is defined.
How can we do? Use only **a part** of the whole support, as we will see later.
A question may arise: is the **error** produced by the missed mass **low enough** to be considered a valid procedure? The following theorem assures that the error is **low** and it helps to **quantify** it!

THEOREM

Suppose that $\widehat{I}_k(\gamma)$ is given by Algorithm 1, and assume that X is finite.

For $\gamma = \dfrac{1}{k\,|X^n|}$ we have, with probability at least $1-\delta$,

$$I(\mu) \;\geq\; (1-\varepsilon_k)\,\widehat{I}_k(\gamma) \;-\; \left(\frac{1}{k} + \big(1 + n\ln(1+k|X|)\big)\varepsilon_k\right), \text{where } \varepsilon_k = \mathbb{E}[U_k] + \sqrt{\frac{\ln(1/\delta)}{k}}.$$

THE ALGORITHM

1: **Input**:

$\mu \in \mathcal{M}_1(\mathcal{X}^n)$ ..... any (pseudo-) joint distribution over $\mathcal{X}^n$

$k \in \mathbb{N}$ ..... sample size

$\gamma \geq 0$ ..... stabilization parameter (typically selected as $1/k$)

2: Independently sample tuples $X_1, \ldots, X_k \sim \mu \in \mathcal{M}_1(\mathcal{X}^n)$

3: Construct a set of indices of unique elements $S = \{i \in [k] \ : \ X_i \neq X_j \quad \forall j < i\}$

4: Construct empirical distributions: for all $i \in S$,

$$\widehat{\mu}(X_i) = \frac{\mu(X_i)}{Z} \ ,$$

$$\widehat{\mu}^{\otimes}(x') = \prod_{i=1}^{n} \frac{\sum_{x^{\setminus i}} \mu(x_1, \ldots, x_{i-1}, x'_i, x_{i+1}, \ldots, x_n)}{Z^{\otimes}} \qquad \text{(for all } x' \in \mathcal{X}^n\text{)}$$

$$\text{where} \quad Z = \sum_{j \in S} \mu(X_j) \ , \quad Z^{\otimes} = \sum_{j \in S} \sum_{x^{\setminus i}} \mu(x_1, \ldots, x_{i-1}, X_{j,i}, x_{i+1}, \ldots, x_n)$$

5: Compute estimate

$$\widehat{I}_k(\gamma) = \sum_{i \in S} \widehat{\mu}(X_i) \ \ln \left(\frac{\widehat{\mu}(X_i) + \gamma/Z}{\widehat{\mu}^{\otimes}(X_i) + \gamma/Z^{\otimes}} \right)$$

EXTRA: ALGORITHM'S VARIATIONS

ALGORITHM 2

Adds **semantic clustering**: merges different but equivalent responses to provide a more coherent MI estimate.

ALGORITHM 3

Reconstructs the pseudo-joint using only LLM queries and applies clustering: the practical version for black-box models.

ABSTENTION POLICIES

How can we score hallucinations?

We can use the uncertainty estimate $\widehat{I}_k(\gamma, x)$ to quantify how much we believe the LLM hallucinates on query X.

→ ***Abstention policies*** are designed based on these scores: the user receives an answer only if the value is below or above a specific threshold.
Calibration is then used to adjust the problem to a specific task.

$$a_\lambda(x) = \begin{cases} 0 & \text{if } I_k(\gamma, x) < \lambda \\ 1 & \text{if } I_k(\gamma, x) \geq \lambda \end{cases}$$

⚠ if $a_\lambda(x)$ is equal to 0, the policy does not abstain and the answer is provided.

EXTRA: FOCUS ON LAMBDA

"MITIGATING LLM HALLUCINATIONS VIA CONFORMAL ABSTENTION", YADKORI ET AL. (2024)

Let X be a space of input prompts and Y a space of output responses.
Let $m : X \times Y' \times Y \to \{0, 1\}$ be the binary ground-truth match function,
working as the indicator of
semantic equivalence of Y' to Y given X.

→ The authors aim to determine whether Y' and Y are **equally correct responses** to X.

GOAL

Find the **abstention threshold** λ that yields the smallest number of abstentions
for a given risk tolerance $\alpha > 0$: $\arg\min_{\lambda \in \Lambda} T(\lambda)$ subject to $R(\lambda) \leq \alpha$.

Since the abstention rate $T(\lambda)$ is a non-decreasing function of λ,
this is equivalent to find the smallest λ for which $R(\lambda) \leq \alpha$.
We denote this optimal threshold by λ^*.

METHODS

Conformal Risk Control (CRC)

We can use the **Conformal Risk Control (CRC)** framework to tune λ. We define the average loss as following:

$$L_n(\lambda) = \frac{1}{n}\sum_{i=1}^{n} \ell(X_i, Y_i; \lambda)$$

$$\text{where } \lambda_n = \inf\{\lambda : \frac{n}{n+1} L_n(\lambda) + \frac{1}{n+1} \leq \alpha\}$$

RCPS

RCPS can be used to obtain the following high-probability result for λ_n computed using conformal risk control:

$$P(\mathbb{E}[\ell(X, Y; \lambda_n) \mid D_n] \leq \alpha + c(\delta, \alpha, n)) \geq 1 - \delta$$

Allowing calibration of λ.

CONCLUSIONS

TEST

- LANGUAGE MODEL : Gemini 1.0 PRO
- MATCH :F1 score with threshold 0.25
- SCORING METHODS:
  - probability of greedy response
  - semantic entropy
  - mutual information score
  - self-verification method

RESULTS

Single labeled dataset :

M.I. and S.E. perform better

Combined datasets :

M.I. overtakes the other methods, as it is able to detect separately epistemic and aleatoric uncertainty.

E.I. abstention rate is very high.

THANK YOU !